

Case Study #3: Forecasting with AR and ARIMA Models

Consider the quarterly data on Walmart revenues (in \$million) from the first quarter (Q1) of 2006 through the third quarter (Q3) of 2025 (the same data set as in case study #2, *673_case2.csv*). The goal is to forecast Walmart's quarterly revenue in the fourth quarter (Q4) of 2025 and Q1-Q4 of 2026-2027.

As you did in *case study #2*, start this case with the following:

- Create time series data set in R using the *ts()* function.
- Develop data partition with the validation partition of 19 periods and the rest for the training partition.

Questions

1. Identify time series predictability.

- a. Using the *AR(1)* model for the entire historical data. Provide and explain the *AR(1)* model summary in your report. Using hypothesis testing, explain if the Walmart revenue is predictable.
- b. Using the first differencing (*lag1*) of the entire historical data and *Acf()* function, provide in the report the autocorrelation plot of the first differencing (*lag1*) with the maximum of 8 lags and explain if Walmart revenue is predictable.

2. Apply the two-level forecast with regression model and AR model for residuals.

- a. For the training data set, use the *tslm()* function to develop a *regression model with quadratic trend and seasonality*. Forecast Walmart's revenue in the validation period with the *forecast()* function (use the associated R code from case #2). *No explanation is required in your report.*
- b. Identify the regression model's residuals for the training period and use the *Acf()* function with a maximum of 8 lags to identify autocorrelation for these residuals. Provide the autocorrelation plot in your report and explain if it would be a good idea to add to your forecast an AR model for residuals.
- c. Develop an *AR(1)* model for the regression residuals, present and explain the model and its equation in your report. Use the *Acf()* function for the residuals of the *AR(1)* model (residuals of residuals), present the autocorrelation chart, and explain it in your report.
- d. Create a two-level forecasting model (regression model with *quadratic trend and seasonality* + *AR(1)* model for residuals) for the validation period. Show in your report a table with the validation data, regression forecast for the validation data, *AR(1)* forecast for the validation data, and combined forecast for the validation period.
- e. Develop a two-level forecast (regression model with *quadratic trend and seasonality* and *AR(1)* model for residuals) for the *entire data set*. Present in your report the autocorrelation chart for the *AR(1)* model's residuals and explain it. Also, provide a data table with the models' forecasts for Walmart revenue in Q4 of 2025 and Q1-Q4 of 2026-2027 (regression model, *AR(1)* for residuals, and two-level combined forecast).

3. Use ARIMA Model and Compare Various Methods.

- a. Use `Arima()` function to fit $ARIMA(1,1,1)(1,1,1)$ model for the *training data set*. Present in your report the summary of this ARIMA model, show and briefly explain the ARIMA model and its equation in your report. Using this model, forecast revenue for the *validation period* and present it in your report.
- b. Use the `auto.arima()` function to develop an ARIMA model using the *training data set*. Present in your report the summary of this ARIMA model, show and explain the ARIMA model and its equation in your report. Use this model to forecast revenue in the *validation period* and present this forecast in your report.
- c. Apply the `accuracy()` function to compare performance measures of the two ARIMA models in 3a and 3b. Present the accuracy measures in your report, compare them and identify, using MAPE and RMSE, the best ARIMA model to apply.
- d. Use the two ARIMA models from 3a and 3b for the entire data set. Present models' summaries in your report. Use these ARIMA models to forecast Walmart revenue in Q4 of 2025 and Q1-Q4 of 2026-2027, and present these forecasts in your report.
- e. Apply the `accuracy()` function to compare the performance measures of the following forecasting models for the *entire data set*: (1) regression model with *quadratic trend and seasonality*; (2) *two-level* model (with $AR(1)$ model for residuals); (3) $ARIMA(1,1,1)(1,1,1)$ model; (4) *auto ARIMA* model; and (5) *seasonal naïve* forecast for the entire data set. Present the accuracy measures in your report, compare them, and identify, using MAPE and RMSE, the best model to use for forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.